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Aim

Find the length of the **Longest Bitonic Subsequence**, which first increases and then decreases.

Algorithm

1. Find the **Longest Increasing Subsequence (LIS)** ending at each element.
2. Find the **Longest Decreasing Subsequence (LDS)** starting from each element.
3. For every element, calculate:
$$\text{LIS}[i] + \text{LDS}[i] - 1$$
4. The maximum value is the answer.

Program:

```
main.py Visualize 
```

```
1 n = int(input())
2 arr = list(map(int, input().split()))
3
4 lis = [1] * n
5 lds = [1] * n
6
7 # Longest Increasing Subsequence
8 for i in range(n):
9     for j in range(i):
10         if arr[j] < arr[i]:
11             lis[i] = max(lis[i], lis[j] + 1)
12
13 # Longest Decreasing Subsequence
14 for i in range(n - 1, -1, -1):
15     for j in range(i + 1, n):
16         if arr[j] < arr[i]:
17             lds[i] = max(lds[i], lds[j] + 1)
18
19 # Longest Bitonic Subsequence
20 answer = 0
21 for i in range(n):
22     answer = max(answer, lis[i] + lds[i] - 1)
23
24 print("Length =", answer)
```

Output:

Output

Length = 4

Time Complexity: $O(n^2)$

Space Complexity: $O(n)$